Trainings

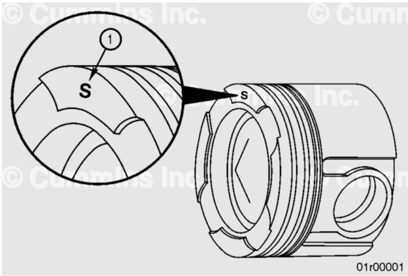
General Information

Two different piston part numbers are used in production on the QSK23 engine. The two different piston part numbers can be indentified by an "L" or and "S" on the head of the piston (1). Pistons marked with a "S" are used for service **only**.

These service pistons can be used to replace a "L" or "S" marked piston. Pistons "L" and "S" can be mixed in the same engine.

Note : Tier 1 and Tier 2 pistons have different part numbers and are not interchangeable.

Note : "S" pistons can be used with both "S" and "L" liners. "L" pistons can only be used with "L" liners.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery first and connect the negative (-) battery last.

⚠ WARNING ⚠

Coolant is toxic; Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

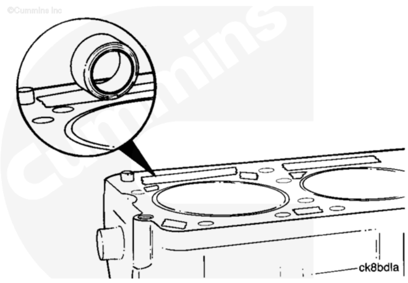
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Disconnect the battery or supply air to the starter. See equipment manufacturer service information.
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html)
- Drain the lubricating oil and remove the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/89/89-007-025.html)
- Remove the oil suction tube and block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
- Remove the intake manifold. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/89/89-010-023.html)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/89/89-011-007.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)

- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/89/89-004-014.html)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-013.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/89/89-002-004-tr.html)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-046.html)

Remove

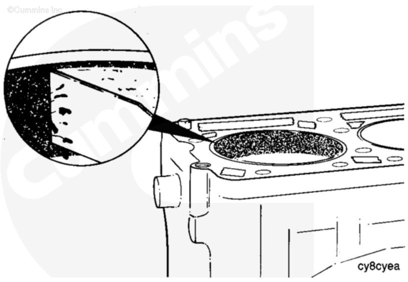
Use tape to protect the push rod galleys, coolant passages, and oil passages from contamination.



⚠ CAUTION ⚠

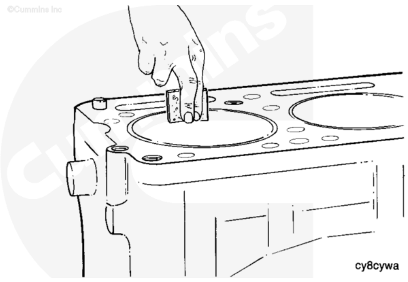
Do not use abrasive paper to remove the carbon deposits. Small particles of abrasive paper will cause severe damage.

Use a scraper or similar blunt-edged tool to loosen the carbon deposits.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility or personal injury.



⚠ CAUTION ⚠

Make sure that no abrasive cleaners or materials are used in the piston ring travel area.

Remove the remaining carbon with an abrasive nylon pad, such as Scotch-Brite™ 7448, and solvent. The carbon **must** be removed, but the surface does **not** have to look like new metal.

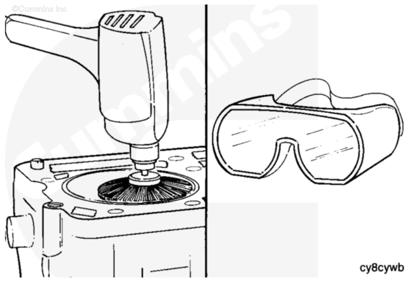
⚠ WARNING ⚠

Wear eye protection to prevent serious eye damage during this operation.

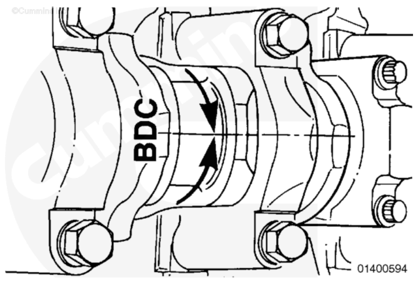
An alternate method to remove the carbon ridge is to use a high-quality steel wire wheel installed in a drill.

Do **not** use a steel wire wheel in the piston ring travel area. Operate the wire wheel in a circular motion to remove the deposits.

Do **not** use a steel wire wheel of inferior quality because it will lose steel bristles during operation and cause additional contamination.

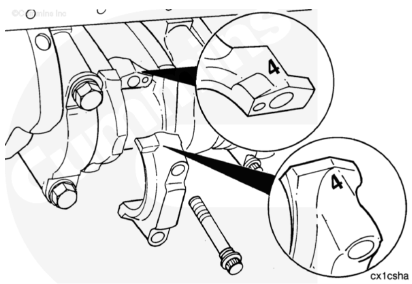


Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at bottom dead center.



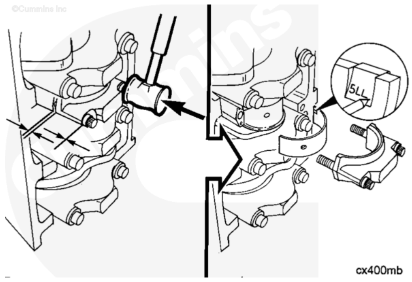
The connecting rods **must** have the cylinder number marked on both the rod and the cap on the side positioned toward the camshaft. Check the rods for correct markings. Use a steel stamp to mark any connecting rod that is **not** correctly marked.

Loosen the capscrews until there is 6 mm [0.236 in] of clearance between the rod cap and capscrew head.

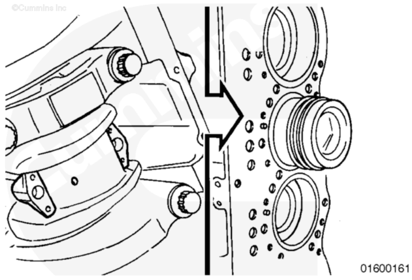


Use a mallet to tap the connecting rod capscrews until the connecting rod cap and connecting rod separate. Remove the capscrews and the connecting rod cap.

Remove the lower rod bearing. Use an awl to mark the bearing position in the tang area for future identification or possible failure analysis.

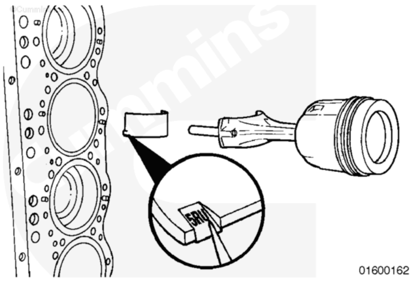


Press the piston and connecting rod up until the piston rings are above the cylinder.



⚠ CAUTION ⚠

Place the piston and connecting rod assembly in a rack to prevent damage to the piston and connecting rod assembly.



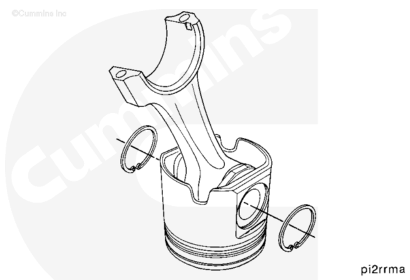
Remove the piston and connecting rod assembly.

Remove the upper connecting rod bearing and place the piston and connecting rod assembly in a rack.

Use an awl to mark the bearing position in the tang area.

Disassemble

Remove the piston rings. Refer to Procedure 001-047 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-047.html)

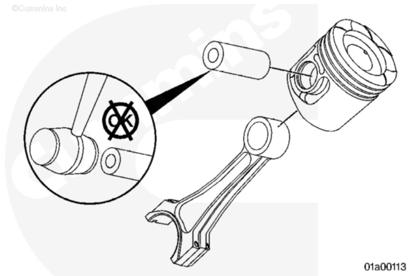


Use snap ring pliers to remove the snap rings from both sides of the piston.

Mark or tag the rings from each cylinder with the cylinder number for analysis, if desired.

⚠ WARNING ⚠

To reduce the possibility of personal injury and damage to the piston, do not drop the piston while removing the piston pin.



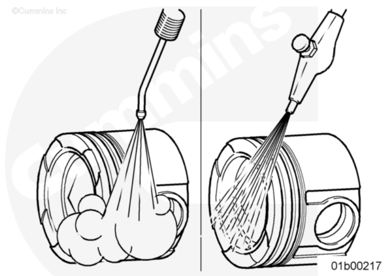
Remove the piston pin from the piston assembly.

Separate the piston and connecting rod.

It is recommended that the piston and connecting rod be kept together for future analysis.

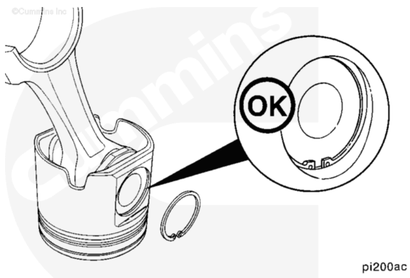
Assemble

Clean and inspect the pistons for reuse. Refer to Procedure 001-043 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-043.html)



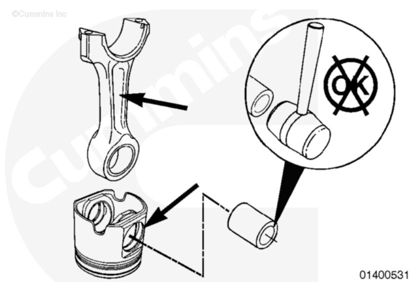
Install a new snap ring in one piston pin bore of each piston.

It is **not** necessary to heat the pistons before assembly. The piston pin is a slip fit.



⚠ CAUTION ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing it to seize in the liner.



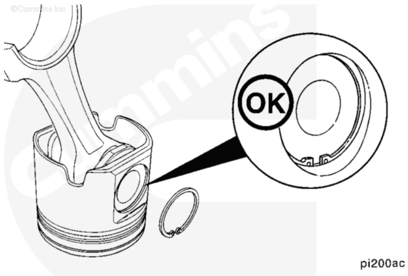
Lubricate the piston pin and connecting rod bushing with clean 15W-40 engine oil.

Align the “X”, stamped on the piston skirt, with the part number cast into the connecting rod and insert the rod into the piston.

Align the pin bore of the connecting rod with the pin bore of the piston and install the piston pin.

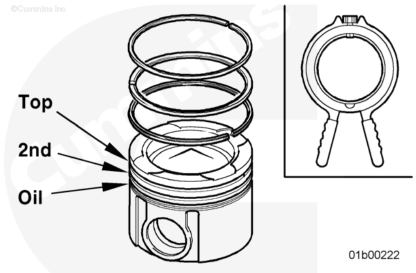
⚠ CAUTION ⚠

The snap ring must be seated completely in the piston groove to reduce the possibility of engine damage during operation.



Install a new snap ring in the piston pin bore.

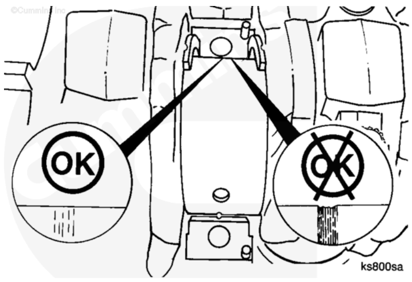
Install the piston rings. Refer to Procedure 001-047 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-047.html)



Clean and Inspect for Reuse

⚠ CAUTION ⚠

To reduce the possibility of damage to the crankshaft, do not use a scraper or a wire brush on the crankshaft journals.

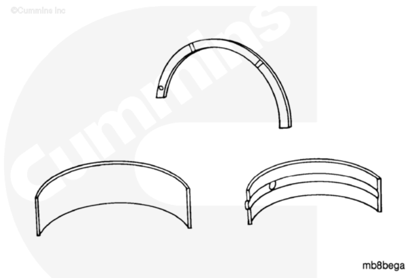


Use a lint-free cloth to clean the crankshaft journal.

Inspect the crankshaft journal for damage.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

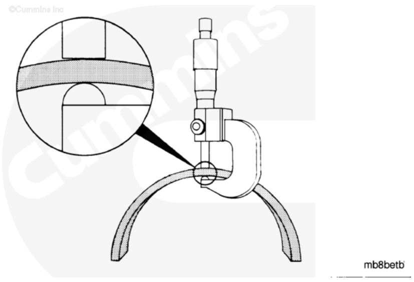
⚠ CAUTION ⚠

To reduce the possibility of bearing damage, do not use a scraper or wire brush on bearing surfaces.

Clean the bearings with solvent and dry with compressed air.

⚠ CAUTION ⚠

To reduce the possibility of damage to the bearing surfaces, use adequate professional tools and measuring techniques. Use of improper measuring tools or methods may cause local surface damages that can become the initiation point for bearing surface layer failure.



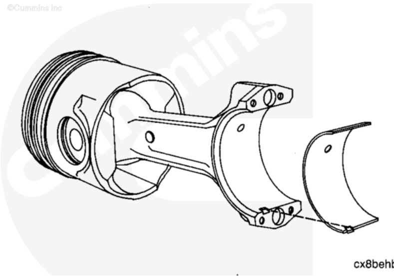
Use a ball-end micrometer to measure the bearing thickness at the wear location.

Connecting Rod Bearing Thickness			
Connecting Rod Bearings	**mm**		**in**
Standard	3.458	MIN	0.1361
	3.471	MAX	0.1367
Oversize 0.25 mm [0.010 in]	3.583	MIN	0.1411
	3.596	MAX	0.1416
Oversize 0.50 mm [0.020 in]	3.708	MIN	0.1460
	3.721	MAX	0.1465
Oversize 0.75 mm [0.030 in]	3.833	MIN	0.1509
	3.846	MAX	0.1514
Oversize 1.0 mm [0.040 in]	3.958	MIN	0.1558
	3.971	MAX	0.1563

Install

Use a lint-free cloth to clean the connecting rod and bearing shells.

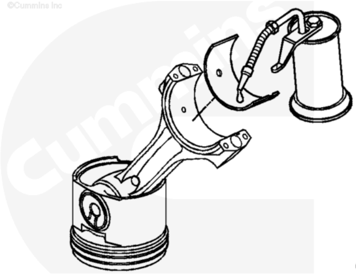
Install the connecting rod bearing. Make sure the tang is positioned in the machined slot as shown in the illustration. The end of the bearing shell **must** be even with the connecting rod cap mounting surface.



cx8behb

⚠ CAUTION ⚠

Do not lubricate the back side of the bearing shells. Engine damage can result.



01400530

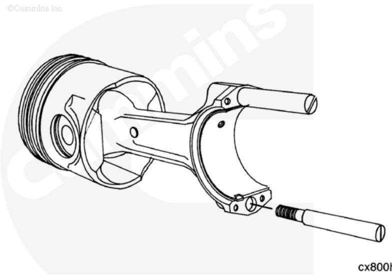
Lubricate the bearing surface with clean 15W-40 engine oil.

The bearings **must** be installed in their original location if new bearings are **not** used.

All of the connecting rod bearing shells are identical.

Install two connecting rod guides, or equivalent, in the connecting rod.

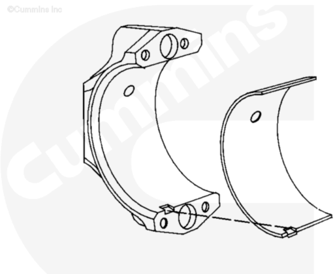
The guide will aid in assembly and protect the crankshaft.



cx800ha

⚠ CAUTION ⚠

The connecting rods and connecting rod caps are not interchangeable. The connecting rods and connecting rod caps are machined as an assembly. Engine failure will result if they are mixed.

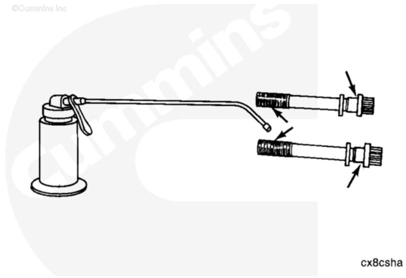


cx8behd

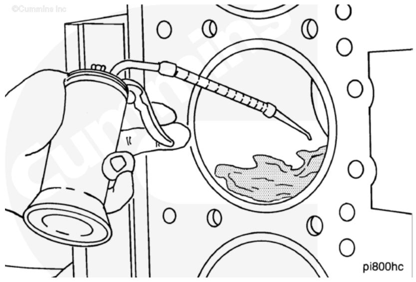
Install the lower bearing shell in the connecting rod cap. Make sure the tang of the bearing shell is in the slot of the connecting rod cap and the end of the bearing is even with the surface of the connecting rod cap.

Lubricate the connecting rod capscrews and washers with clean 15W-40 engine oil.

Install the washers and capscrews in the connecting rod cap.

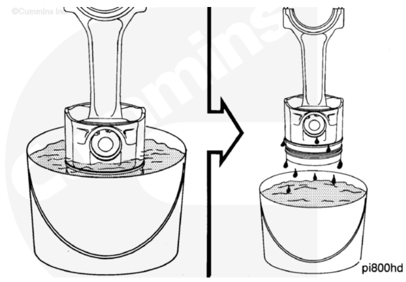


Lubricate the cylinder liner with clean engine oil. The entire liner bore **must** be lubricated.

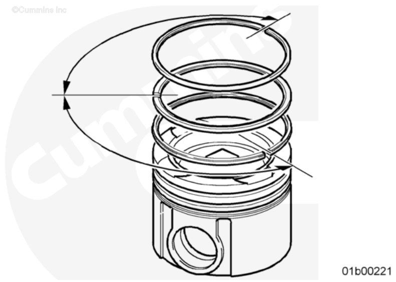


Immerse the piston in clean engine oil until the piston rings are covered.

Allow the excess oil to drip off the assembly.

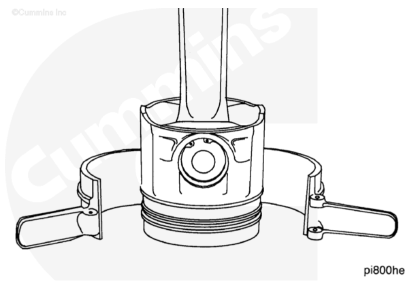


Make sure the ring gap position is still correct. Refer to Procedure 001-047 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-047.html)



⚠ CAUTION ⚠

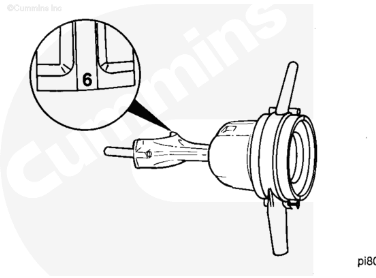
Make sure the rings fit correctly in the piston ring grooves or engine damage can result.



Use piston ring compressor, Part Number 3164459, or equivalent, to install the ring compressor on the piston.

⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the connecting rod and connecting rod cap must be the same. Damage to the engine can occur if the connecting rods and connecting rod caps are mismatched.



pi800hf

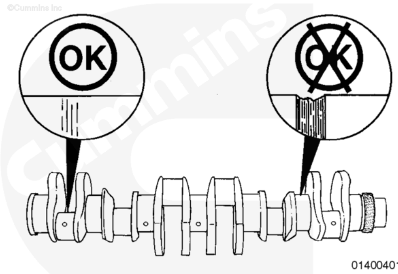
⚠ CAUTION ⚠

The “X” stamped on the piston skirt and the part number, cast into the connecting rod, must be facing the front of the engine. If the connecting rods are installed backward, engine damage can occur.

Rotate the crankshaft until the journal for the connecting rod to be installed is at top dead center and centered in the cylinder bore.

Note : The crankshaft is illustrated out of the engine for clarity.

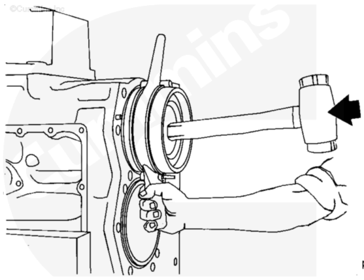
Inspect the crankshaft rod journals for damage.



01400401

⚠ CAUTION ⚠

The “X” stamped on the piston skirt and the part number, cast into the connecting rod, must be facing the front of the engine. If the connecting rods are installed backward, engine damage can occur.



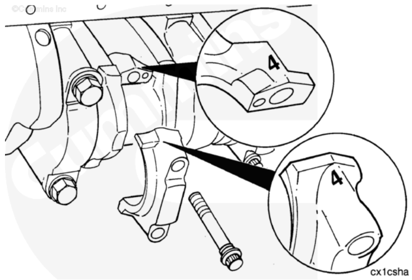
pi400ha

Align the part number, cast into the connecting rod, toward the front of the engine. Install the connecting rod and piston into the cylinder bore until the ring compressor touches the cylinder block. Align the rod with the crankshaft journal.

Hold the ring compressor firmly against the block. Use a wooden hammer handle to push the piston into the liner.

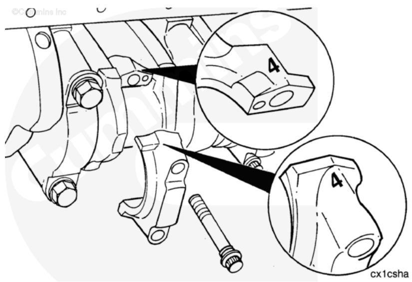
⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the connecting rod and connecting rod cap must be the same. Damage to the engine can occur if the connecting rods and connecting rod caps are mismatched.



Remove the guide pins and install the connecting rod cap.

Note : Check the number of punch marks on the head of the mounting capscrews. If there are four punch marks, do not reuse the capscrew. Use a new capscrew.



Use clean engine oil to lubricate the connecting rod capscrews.

Assemble the connecting rod, cap, and capscrews.

Always use the torque-turn method to tighten the connecting rod capscrews.

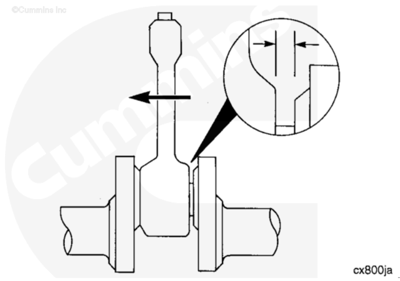
Use the following steps to tighten the capscrews alternately and evenly.

Torque Value:

1. 196 n•m [145 ft-lb]
2. Tighten 90°

Note : After tightening the capscrews, make one punch mark on the head of each capscrew. Do not make a punch mark when using a new capscrew.

Check the side clearance between the connecting rod and the crankshaft.



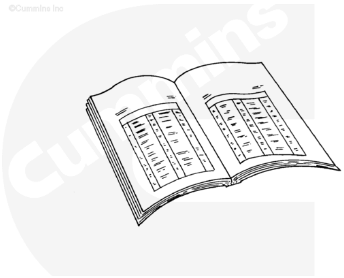
The connecting rod **must** move freely from side-to-side.

Connecting Rod and Crankshaft Side Clearance New or Remanufactured Parts		
mm		in
0.200	MIN	0.008
0.374	MAX	0.015

Finishing Steps

⚠ **WARNING** ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery first and connect the negative (-) battery last.



- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-046.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/89/89-002-004-tr.html)
- Install the rocker lever housing. Refer to Procedure 003-013 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-013.html)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/89/89-004-014.html)
- Install the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)
- Adjust the overhead set. Refer to Procedure 003-006 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-006.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/89/89-003-011.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/89/89-011-007.html)
- Install the intake manifold. Refer to Procedure 010-023 in Section 10. (/qs3/pubsys2/xml/en/procedures/89/89-010-023.html)
- Install the oil suction tube and block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/89/89-007-025.html)
- Fill the engine with clean 15W-40 lubricating oil. Use the following procedure for engine oil capacity. Refer to

Procedure 007-037 in Section 7 (/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)

- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html).
- Connect the battery or supply air to the starter. See equipment manufacturer service information.
- Start the engine. Check for proper operation.